

ALPACA TRADING BOT HACKATHON 2026



ELITE-BOT

Autonomous AI Multi-Asset Trading Platform

Bridging quantitative statistical arbitrage, multi-agent LLM consensus (CrewAI), and real-time risk guards — all executed through the Alpaca Markets API.

The Problem: Why 90% of Retail Trading Bots Fail



The 'Blind Math' Trap

- Pure quantitative indicators (Z-score, RSI, MACD) assume stationary historical price behavior.
- When real catalysts strike — earnings surprises, CEO scandals, FTC lawsuits — pairs diverge permanently.
- Retail bots blindly average down into 'value traps' and blow up accounts.



Primitive Risk Management

- Most bots only support static stop-losses configured once, at script startup.
- No real-time awareness of earnings blackout windows, overnight gaps, or volatile opens.
- Users lack instant controls to adjust stop-loss limits or kill live positions.



Black-Box CLI Disconnect

- Algorithmic trading stays locked behind dense terminal scripts with no visualization.
- Key security risk: live broker API credentials stored in cleartext disk files.
- Zero visibility into why a trade was accepted or rejected in real time.

Our Solution: The Tri-Layer Autonomous Architecture

LAYER 1



Quant Brain

Mathematical precision

- Kalman filter dynamic hedge ratios
- GARCH(1,1) volatility forecasting
- Hidden Markov Model regime detection
- Cointegration & dynamic Z-score scanner
- Fractional Kelly Criterion position sizing

LAYER 2



Multi-Agent AI

Qualitative committee (CrewAI)

- 4 autonomous LLM agents (via Groq & OpenRouter)
- Senior Quant Analyst · News & Catalyst Scout
- Bearish Contrarian Risk Officer · Chief Investment Officer
- Filters out ~65% of false traps

LAYER 3



Alpaca Execution

Broker integration

- Ultra-fast paper & live order execution
- Dynamic hard & soft stop-loss controls
- 24/7 autonomous position manager
- Safe market hours & earnings blackouts
- Interactive React + FastAPI dashboard

Deep Alpaca Integration: Powering Seamless Execution



Dual Paper & Live Trading Engines

- Instant one-click toggle between Alpaca Paper and Live environments.
- Dynamically re-binds endpoints (paper-api vs. api.alpaca.markets).
- Zero server downtime or script restarts required.



Real-Time Account & Position Sync

- Live WebSocket/REST sync of equity, cash, and buying power.
- Continuous PnL calculation across long/short spread legs.
- Instant single-click 'Close All Positions' panic button.



High-Speed Order Execution & Sizing

- Automated bracket, market, and limit order routing for equities & crypto.
- Fractional Kelly sizing checks portfolio equity to prevent margin calls.
- Strict fee-aware calculations for slippage & commission.



Bank-Grade Key Isolation & Security

- Zero plain-text API keys on disk — held in browser memory & runtime env.
- Key masking in the UI (PK12****ABCD) for secure demos.
- Dynamic module hot-reloading refreshes credentials every session.

Multi-Agent AI Validation: The Digital Investment Committee



1

Senior Quant

Validates mathematical spread integrity.

KEY CHECKS

- Z-score divergence
- Kalman hedge stability
- Volume anomaly verification
- Mean-reversion probability



2

News & Catalysts

Real-time macro & corporate catalyst scanner.

KEY CHECKS

- Breaking SEC filings
- Earnings blackout windows
- Sector sentiment shifts
- M&A / lawsuit news



3

Bear Risk Officer

Actively tries to disprove and kill the trade.

KEY CHECKS

- Value-trap detection
- Correlation breakdown risk
- Asymmetric downside
- Liquidity / slippage hazards



4

Chief Investment Officer

Synthesizes all reports into the final decision.

KEY CHECKS

- Strict JSON output
- Confidence score (0–100)
- Signal strength (0–1.0)
- Action: BUY / SELL / SKIP

Risk Protocol: Multi-Layer Capital Protection



Configurable Hard & Soft Stop Loss

- Manual UI control for both dollar (\$) and percentage (%) stop losses.
- Soft Stop (-\$75): automatically reduces position exposure by 50%.
- Hard Stop (-\$100): instant full liquidation kill switch via Alpaca API.



Autonomous Background Trade Guardian

- Dedicated daemon thread evaluates all open positions every 60 seconds.
- Asks: 'If we didn't hold this position now, would we enter it?'
- Triggers AI-guided exits to lock profits or prevent drawdowns.



Safe Market Hours & Earnings Blackout

- Market Hours Guard: trades only between 9:35 AM – 3:45 PM ET.
- Earnings Blackout: skips tickers with earnings within 5 calendar days.
- Overnight Gap Protection: blocks new entries after 3:30 PM ET.



Real-Time Global Ticker Freeze

- Instant blacklist via the UI for any problematic or meme stocks.
- Immediately recognized across all trading engines (Elite, Crypto, Smart).
- Sector Exposure Guard caps max allocation per industry at 25%.

Modern Full-Stack Dashboard (React 19 + FastAPI)



Live Mission Control

- Real-time metrics: portfolio value, unrealized PnL, win rate, active bots.
- Streaming log interceptor — zero-latency terminal feed with color-coded severity.
- Instant one-click start/stop bot engines & emergency panic liquidation.



Dynamic Risk Customization

- Hard stop-loss input (\$ and %) · Soft stop-loss & profit targets.
- Max hold time (hours) · Max sector exposure slider.
- Global ticker freeze / unfreeze tag manager.
- Goal-seeking urgency engine for capital targets.



Backtester & Analytics

- Historical pair backtesting — multi-year statistical arbitrage with transaction costs.
- Interactive charts with Z-score bands and entry/exit markers.
- Quant health overview: real-time Kalman & GARCH stability scores.

Multi-Asset & Multi-Strategy Bot Portfolio



Elite Trader

Alpaca US Equities

Pairs trading with Kalman Filter + GARCH + 4-agent CrewAI consensus. Our flagship, hedge-fund-grade engine.



Crypto Elite

Alpaca Crypto

High-frequency crypto spread arbitrage with a reinforcement-learning (Q-Learning) brain. Trades BTC, ETH, SOL — 24/7.



Smart Momentum

Alpaca US Equities

Single-ticker momentum & breakout scanner combining RSI divergence with AI catalyst validation.

Competitive Matrix: Why Elite-Bot Stands Out

Feature / Capability	Standard Retail Bots	Traditional Quant Funds	Elite-Bot (Our Project)
Qualitative News & Catalyst Filter	None (pure math)	Expensive manual teams	Autonomous multi-agent LLMs
Execution Speed & Ease	Clunky terminal scripts	Proprietary private infra	1-click Alpaca REST & WebSockets
Risk Controls & Hard Stop Loss	Static config only	Proprietary risk desks	Dynamic UI stop loss & 24/7 guardian
Open-Source & 1-Click Deploy	Fragmented codebases	Closed source / \$1M+ fee	Free deploy to Render & Docker

● Strong advantage ● Partial / limited ● Not available

Summary: The Future of Autonomous Alpaca Trading



Key Hackathon Highlights

- Deepest utilization of Alpaca's API (Paper/Live, Positions, Account, Crypto).
- Novel synthesis of high-end quant math + multi-agent LLM reasoning.
- Zero-compromise security architecture — no keys on disk, isolated instances.



Ready to Deploy Today

- 100% working full-stack system (React 19 + FastAPI + CrewAI + Alpaca).
- 1-click deploy to Render with Docker support.
- Tested with live Alpaca paper trading accounts.



Future Roadmap

- Alpaca Options integration (volatility arbitrage & delta-neutral straddles).
- Reinforcement-learning fine-tuning on historical Alpaca fills.
- Mobile app companion with push alerts for trade decisions.



Try It Out

- GitHub: github.com/rudrakshmala/elite-bot
- Powered by Alpaca Markets & Groq.
- Thank you, judges and hackathon organizers!